

WANG MODEL 700A/B ELECTRONIC CALCULATOR



DESCRIPTION

The WANG 700A/B is the most advanced programmable electronic calculator on today's market. The following units are included in one self-contained desk-top unit:

Display, Keyboard, Read-only Memory (ROM), Core Memory, Tape Drive.

DISPLAY

Two rows (X & Y) are displayed.

Type: 1/2 inch display tubes.

Capacity: 12 digits, sign & decimal, 2-digit exponent and sign.

Dynamic Range: $\pm 10^{-98}$ to $\pm 10^{+99}$

Decimal Setting: Automatic floating decimal point from 0 to $\pm 10^{+10}$.

Automatically converted to fixed point from $\pm 10^{+11}$ to $\pm 10^{+98}$.

KEYBOARD

Keyboard: Easily replaceable microswitches with color coded key tops.

Data Entry: Standard 10-key keyboard plus decimal. Exponent can be entered separately. Shifting and exchange between X & Y registers.

Arithmetic: +, -, $\times$, $\div$, X^2 , $\sqrt{x}$, $1/X$

Logarithmic: $\text{LOG}_e X$, $\text{LOG}_{10} X$

Exponential: e^x , 10^x

Miscellaneous: $|X|$, π , Integer of X

Special Functions: 16 changeable special function keys by means of furnished tape programs.

Standard TRIGONOMETRIC:

DEGREES TO RADIANS, RADIANS TO DEGREES, $\text{SIN}(X)$, $\text{COS}(X)$, $\text{TAN}(X)$, $\text{SIN}^{-1}(X)$,

Standard TRIGONOMETRIC (Cont.):

$\text{COS}^{-1}(X)$, $\text{TAN}^{-1}(X)$, RECTANGULAR TO POLAR, POLAR TO RECTANGULAR, $\text{SINH}(X)$, $\text{COSH}(X)$, $\text{TANH}(X)$, $\text{SINH}^{-1}(X)$, $\text{COSH}^{-1}(X)$, $\text{TANH}^{-1}(X)$.

Standard STATISTICS:

x , σ , σ^2 , (ungrouped), x , σ , σ^2 (grouped)

NORMAL DISTRIBUTION, INVERSE NORMAL DISTRIBUTION, X^2 STATISTIC, X^2 DISTRIBUTION, ERROR FUNCTION, BINOMIAL DISTRIBUTION, $N!$, LINEAR REGRESSION, GAMMA FUNCTION, NEGATIVE BINOMIAL DISTRIBUTION, POISSON DISTRIBUTION, RANDOM NUMBER GENERATOR.

Other Functions: Available as programming packages (see price list).

SPEEDS

Function	Approximate Speed
Addition/Subtraction	300 to 500 microseconds
Multiplication/Division, X^2 , $1/X$	2 to 5 milliseconds
10^x , $\text{Log}_{10} X$	15 milliseconds
$\sqrt{x}$	25 milliseconds
e^x , $\text{Log}_e X$	20 milliseconds
Single Variable Trig Functions	135 to 250 milliseconds
Read/write cycle time	2 microseconds

DOUBLE PRECISION (RECALL RESIDUE)

Up to 24-digit accuracy can be gained after +, -, $\times$. Remainder is displayed after $\div$.

MEMORY

The Wang 700A/B contains 8,192 bits of core memory. This is organized as 120 data storage registers; or, as storage for 960 program steps.

Example

100 Steps plus	106 Registers
250 Steps plus	88 Registers
500 Steps plus	56 Registers
750 Steps plus	26 Registers

ADDRESSING STORAGE REGISTERS DIRECT ADDRESSING

(by storage keys and toggle switches)

Operations: Store, recall, or  is performed in two steps.

Internal Arithmetic: +, -, X, ÷ is performed in two steps in all 120 registers.

INDIRECT ADDRESSING

Register Address is indicated in the Y-register. It can be listed in the program or manually keyed in.

Operations: Store, recall, or  is performed in one step.

Internal Arithmetic: +, -, X, or ÷ is performed in one step in all 120 registers.

Subroutines: 64 different subroutines can be addressed manually or programmed from any point in a program by one step using the toggle switches. At the end of a subroutine, a **RETURN** branches back. Two levels of subroutines are possible on the 700A, 5 levels on the 700B.

Step and Display Program: Simultaneously, step and code numbers are displayed for checking and changing individual steps on a random access.

Verify Program: This command accumulates all code numbers of a program. The total is displayed and verifies to the user that he has properly loaded the correct program — also used to by-pass programs.

Recording: A **RECORD PROGRAM** command can record programs from core memory onto a tape cassette.

PROGRAMMING

Method: Entered via keyboard into core memory by pressing keys in proper sequence (**LEARN** mode) or by loading of a magnetic tape cassette.

Capacity: In core = 960 steps; on one cassette, 23,000 steps.

Automatic Loading: Allows execution of extremely long programs.

Protection: Each of the two sides of a cassette can be individually protected as desired.

Commands: A total of 95 different instructions are available from the keyboard.

Unconditional Branching: A **SEARCH XX** statement can be programmed to branch to the corresponding **MARK XX** in a program.

Conditional Branching: **SKIP IF:** $Y < X$, $Y = X$, $Y \geq X$, $Y = 0$, $Y \neq 0$, $X = 0$, $X \neq 0$, Y positive, Y negative, X positive, and X negative are commands used for conditional branching.

Test for Error: **SKIP IF ERROR** tests for illegal mathematical operation, overflow or non-existing address (see next paragraph).

Error Lights: **MACHINE ERROR:** indicates improper loading of tape. **PROGRAM ERROR:** indicates illegal operations (see previous paragraph).

PULL-OUT INSTRUCTIONS

Condensed operating instructions on durable swing-out plastic plates are mounted under the keyboard for quick reference.

PERIPHERALS

Model 701 Alpha/numeric Output Writer, Model 702 Plotting Output Writer (requires 700B), Model 703 Paper Tape Editor, and a Model 704 Card Reader are available (see separate literature).

INPUT/OUTPUT CAPABILITY

An input/output connector located on the rear panel of every Model 700 Series calculator, enables direct interfacing of various external equipment. Through this connector, data information can be received, program operations can be executed, or program steps can be "learned" directly into the core memory. Output information, general data, keyboard executions, and the contents of the X-register display are also available.

Specifications describing the signals and signal functions required and/or appearing at the input/output connector are available upon request.

GENERAL

Dimensions: 8.5 in. high, 20 in. wide, 20.5 in. deep

Weight: Net 35 lbs., gross 58 lbs.

Power: 115 vac $\pm 10\%$, 60 hz, 60 watts (standard)
230 vac $\pm 10\%$, 50 Hz, 60 watts (optional,
no charge)

WARRANTY

Full warranty of 90 days for equipment: one year for parts only. Optional maintenance contracts available.